83/100

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1521

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Duration: 60 Minutes** **Total Marks: 25**

Question Count: 5

Code of Conduct

I V.C.Bharan Baisav 192511369 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.C.Bharan Baisav

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty ....**Total Marks Awarded:**...................

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO1-A71 - SHORT ANSWERS TEST
CSA1521 - CLOUD COMPUTING

V.C. Bhavan Baisw
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1) Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.

A: Cloud computing evolved over several decades to overcome the limitations of standalone systems such as high cost, limited resources, poor scalability.

Evolution of computing:

1) Standalone computing:
- A single computer performs all tasks.
- Limited processing power and storage.
- Example: Personal desktop computer.

2) Parallel computing:
- Multiple processors in one computer execute tasks simultaneously
- Improves processing speed.
- Eg: Scientific simulations.

3) Distributed computing:
- Multiple computers connected through a network share workloads.
- Improves scalability and fault tolerance
- Eg: Google search servers.

4) Grid computing:
- Resources from different organizations work together as a virtual supercomputer

- used for large research projects.
- Eg: CERN Research.

5) Service oriented computing (soc):
- Software functions are provided as reusable services.
- Technologies include SOAP, REST, XML & WSDL

6) Cloud computing:
- Provides computing resources over the internet on demand using virtualization.
- Supports pay-as-you-go pricing and rapid scalability.

Key Technological Transitions:
- Multi core processors.
- High-speed networking.
- Virtualization.
- Web services.
- Internet technologies
- Data centers

Example:

Amazon Web Services (AWS) provides virtual servers, storage & networking on demand without users purchasing physical hardware

2) Interpret the working of hypervisors in virtualization. Differentiate between type 1 and type 2 hypervisors with examples.

A hypervisior is software that creates and manages virtual Machines (VMs) on a single physical computer. It allocates CPU, memory, storage and networking resources to each VM.

Working of Hypervisor:

- Physical hardware is installed.
- Hypervisor is loaded.
- Multiple virtual machines are created.
- Each VM runs its own operating system independently.
- Resources are shared efficiently among all VMs.

FEATURE	TYPE-1-HYPERVISOR	TYPE-2-HYPERVISOR
Runs On	Physical hardware	Host operating system
Performance	High	Moderate
Security	Better	Lower
Usage	Data centers and cloud	Personal computers
Examples	VMware ESXi, Microsoft Hyper-V, Xen	Oracle VirtualBox, VMware Workstation

Example:

AWS uses Type-1-hypervisors to run thousands of Virtual Machines securely on physical servers.

3) Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services.

An application programming Interface (API) is a set of rules that allows different software applications to communicate and exchange data.

Role of APIs in cloud Integration:

- Connect different cloud services.
- Enable communication between applications.
- Automate cloud operations.
- Allow developers to access cloud resources programmatically.

APIs support interoperability by:

- Using standard protocols such as HTTP and HTTPS.
- Exchanging data in JSON or XML format.
- Allowing applications developed in different programming languages to communicate
- Integrating services from different cloud providers.

Example:

An online shopping website uses APIs to connect with:

- payment gateway.
- Email notification service.
- cloud Database
- Delivery tracking service.

Although these services are built using different technologies, APIs enable seamless communication.

4) Explain the concept of resource pooling in cloud computing. Illustrate how it benefits the multiple users.

Resource pooling is a cloud computing feature where computing resources such as servers, storage and networking are shared among multiple users using virtualization.

Working:

- cloud providers maintain a pool of resources.
- Resources are dynamically allocated based on user demand.
- Users and share the same infrastructure but remain isolated from each other.

Benefits:

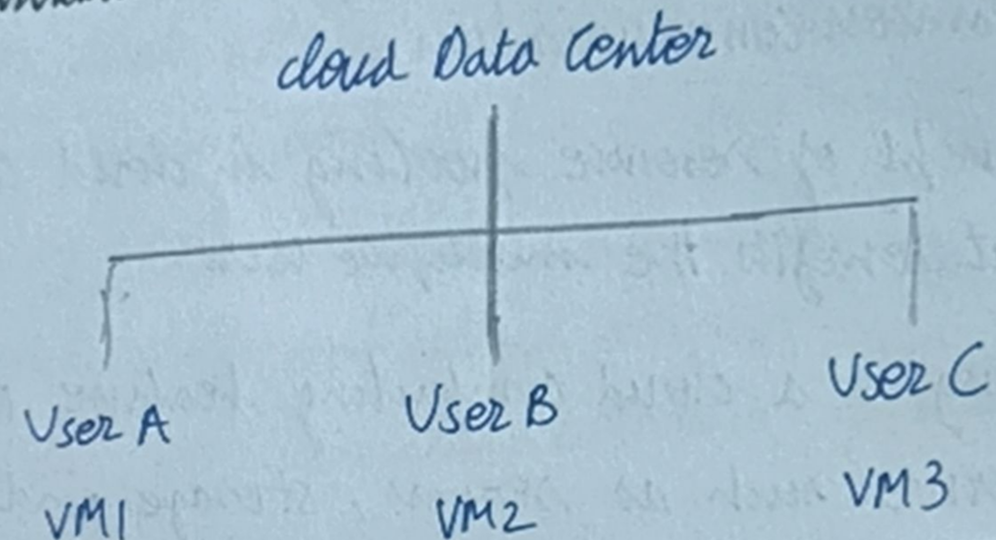
- Better Resource utilization.
- Reduced infrastructure cost
- High Scalability.

- Improved availability.
- Efficient workload management.

Example:

AWS uses large data centers where many customers share the same physical servers through virtualization while each customer receives independent virtual machines.

Illustration:



5) Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Rapid elasticity is the ability of cloud computing to automatically increase or decrease computing resources according to workload demand.

Working:

- Resources are monitored continuously.
- When demand increases, additional servers or virtual machines are allocated.
- When demand decreases, unused resources are released automatically.

Benefits:

- Handles sudden traffic spikes.
- Reduces operational cost.
- Prevents resource wastage.
- Improves application performance.
- Ensures high availability.

Example:

During an online shopping sale (Amazon or Flipkart), cloud servers automatically increase to support millions of users. After the sale ends, the extra servers are removed.

Dynamic Workloads:

- Video streaming
- online Examinations
- E-commerce Websites
- Banking applications
- Social media platforms.